

Literature-Implied Answer to a Vector-Valued Null Universal Differentiability Question

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Status

This is a lightweight literature-implied answer packet. It records that a finite-dimensional vector-valued question mentioned in the introduction of Doré–Maleva [1] is answered by later work of Preiss–Speight [2], together with the Alberti–Csörnyei–Preiss and Csörnyei–Jones results cited there. This is not a new proof produced by the run.

Source Question

Doré–Maleva [1] prove that every nonzero Banach space with separable dual contains a closed bounded scalar universal differentiability set of Hausdorff dimension 1. In the introduction, they also discuss the finite-dimensional vector-valued analogue. They ask whether there are Lebesgue null sets in $\mathbb{R}^n$ containing a differentiability point of every Lipschitz map

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

when $n \geq m \geq 2$. The source says that the answer was then known only for $m = 2$, that the case $n = m = 2$ is negative, and that no similar positive results were known when the codomain dimension is at least 3.

Supporting Answer

Preiss–Speight [2] state in their introduction that the converse to Rademacher’s theorem is valid if and only if $m \geq n$. Equivalently:

- If $m \geq n$, every Lebesgue null $N \subset \mathbb{R}^n$ is contained in the non-differentiability set of some Lipschitz map $\mathbb{R}^n \rightarrow \mathbb{R}^m$. Thus no Lebesgue null universal differentiability set exists in this range.
- If $n > m$, the converse fails. Hence there is a Lebesgue null $N \subset \mathbb{R}^n$ such that every Lipschitz $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at some point of N .

Theorem 1 of [2] proves the endpoint positive case $m = n - 1$: for every $n > 1$, there is a Lebesgue null set

$$N \subset \mathbb{R}^n$$

containing a differentiability point of every Lipschitz

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^{n-1}.$$

The same introduction says that, by a simple modification explained later in the paper, for every $n > m$ and every $\tau > 0$ there is such an N of Hausdorff dimension at most $m + \tau$.

Identification

The Doré–Maleva question is exactly the finite-dimensional null universal differentiability set question for vector-valued Lipschitz maps. The Preiss–Speight theorem and cited converse results settle the dichotomy: positive precisely for $n > m$, negative for $m \geq n$. Therefore the question should not be treated as a live target for new proof work in this run.

Search and Scope Notes

The local run indexes had no exact packet for arXiv:1103.5094. The local source corpus was searched for the arXiv id, the source title, “universal differentiability set”, “Lebesgue null”, and the n, m vector-valued phrasing. The relation to [2] was found in local parsed sources: arXiv:1304.6916 explicitly says that the converse to Rademacher’s theorem is completely answered, and cites Doré–Maleva for the scalar $m = 1$ UDS developments. The supporting paper does not appear to be written as an answer to this exact paragraph of [1], so the packet is classified as literature-implied rather than literature-already-answered.

References

- [1] M. Doré and O. Maleva, *A universal differentiability set in Banach spaces with separable dual*, arXiv:1103.5094; J. Funct. Anal. 261 (2011), 1674–1710.
- [2] D. Preiss and G. Speight, *Differentiability of Lipschitz Functions in Lebesgue Null Sets*, arXiv:1304.6916.